Shadow obstruction towers and the algebraicity of chiral deformation invariants

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Abstract

For a chirally Koszul vertex algebra A with modular characteristic κ, cubic shadow α, and quartic shadow S_4, we prove that the infinite sequence of higher deformation invariants $\{S_r\}_{r\geq 2}$ extracted from the bar construction is algebraic of degree 2: the weighted generating function $H(t) = \sum_{r\geq 2} r\, S_r\, t^r$ satisfies $H(t)^2 = t^4\, Q(t)$, where

$$Q(t) \;=\; 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$$

is the *shadow metric*, a quadratic polynomial determined by the genus-0 OPE data of A. The critical discriminant $\Delta = 8\kappa S_4$ partitions all chirally Koszul algebras into four classes: Gaussian ($\Delta = 0$, $\alpha = 0$; depth 2), Lie ($\Delta = 0$, $\alpha \neq 0$; depth 3), contact (depth 4, by stratum separation), and mixed ($\Delta \neq 0$; depth ∞). The entire infinite tail $\{S_r\}_{r\geq 5}$ is determined by the three invariants (κ, α, S_4). We compute shadow obstruction towers explicitly for all standard families (Heisenberg, affine Kac–Moody, $\beta\gamma$, Virasoro, $\mathcal{W}_N$), derive the shadow connection and its Koszul monodromy -1, and extract the genus-2 cross-channel correction $\delta F_2(\mathcal{W}_3) = (c + 204)/(16c)$ (the multi-generator universality problem is resolved negatively: this correction is nonvanishing; see Remark 7.1) by a graph-by-graph stable graph sum over the seven genus-2 stable graphs.

Remark 0.1 (Relationship to companion papers). The c-independent identity $S_3(\mathrm{Vir}_c) = 2$ is now proved as a theorem; see [14]. The seven-face programme connecting the shadow tower to Gaudin models, Sklyanin brackets, and GZ26 commuting Hamiltonians is developed in [15].

1 Introduction

1.1 Vertex algebras and moduli of curves

A vertex algebra A on a smooth projective curve X produces, for each stable pair (g, n), a space of conformal blocks $\mathbb{V}_{g,n}(A)$ on $\overline{\mathcal{M}}_{g,n}$. When A is chirally Koszul (Definition 2.4), computing these spaces reduces to the operator product expansion (OPE).

Sewing and factorization express genus-g amplitudes as sums over stable graphs Γ: each vertex contributes a genus-g_v conformal block, each edge a propagator. These graph sums are the Feynman expansion of a two-dimensional conformal field theory and produce characteristic classes on $\overline{\mathcal{M}}_{g,n}$ whose integrals are the free energies $F_g(A)$.

We prove that for any chirally Koszul vertex algebra, the full tower of deformation invariants underlying these free energies is controlled by a single algebraic relation of degree 2.

1.2 The bar construction and the shadow obstruction tower

The algebraic framework is the chiral bar construction. For a vertex algebra A on a curve X, the bar complex $B(A)$ is a factorization coalgebra on $\mathrm{Ran}(X)$ whose differential encodes OPE collision residues on the Fulton–MacPherson compactifications $\mathrm{FM}_n(X)$. When A is *chirally Koszul* (bar cohomology $H^\bullet(B(A))$ concentrated in bar degree 1), the bar complex carries a Maurer–Cartan element

$$\Theta_A \;:=\; D_A - d_0 \;\in\; \mathrm{MC}\big(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)\big),$$

where D_A is the full bar differential (incorporating contributions from all genera and all boundary strata of $\overline{\mathcal{M}}_{g,n}$) and d_0 is the genus-0 linear part. The equation $D_A^2 = 0$, which is a formal consequence of the boundary-of-boundary relation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$, ensures that Θ_A satisfies the Maurer–Cartan equation

$$d_0\,\Theta_A + \tfrac{1}{2}[\Theta_A, \Theta_A] = 0$$

in the modular convolution algebra $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)$. This single equation packages all genera and all degrees simultaneously.

The *shadow obstruction tower* of A consists of the finite-order projections

$$\Theta_A^{\le 2}, \quad \Theta_A^{\le 3}, \quad \Theta_A^{\le 4}, \quad \ldots$$

retaining components of degree at most r. On a one-dimensional primary line $L \subset \mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)$ spanned by a generator η of conformal weight h, the degree-r projection reduces to a scalar $S_r = S_r(A) \in \mathbb{Q}(c)$:

- $S_2 = \kappa(A)$, the *modular characteristic*: the genus-1 curvature scalar determining $F_1 = \kappa/24$.
- $S_3 = \alpha(A)$, the *cubic shadow*: the three-point collision residue on $\mathrm{FM}_3(X)$.
- $S_4 = S_4(A)$, the *quartic shadow*: the four-point collision residue, encoding the first genuinely nonlinear OPE invariant.
- S_r for $r \ge 5$: the higher shadows.

The main question is: *how much of the OPE is needed to determine the full tower?*

1.3 Main results

The answer is *three invariants*: the entire infinite tower $\{S_r\}_{r\ge 2}$ is algebraic of degree 2, determined by the genus-0 triple (κ, α, S_4).

Theorem 1.1 (Riccati algebraicity; Theorem 4.1). *Let A be a chirally Koszul vertex algebra, L a primary line with shadow data S_r, and $\kappa = S_2 \neq 0$. Define*

$$Q(t) \;=\; 4\kappa^2 + 12\kappa\alpha\, t + (9\alpha^2 + 16\kappa S_4)\, t^2$$

and $H(t) = \sum_{r\ge 2} r\, S_r\, t^r$. Then

$$H(t)^2 \;=\; t^4\, Q(t).$$

Every shadow coefficient S_r with $r \ge 5$ is determined by (κ, α, S_4) via the closed form $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q(t)}$.

An immediate structural consequence. The *critical discriminant* $\Delta := 8\kappa S_4$ gives the Gaussian decomposition

$$Q(t) \;=\; (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2.$$

The vanishing or nonvanishing of Δ controls the depth of the shadow obstruction tower:

Theorem 1.2 (Four-class partition; Theorem 5.1). *On any primary line L with $\kappa \neq 0$, the shadow depth $r_{\max} := \sup\{r : S_r \neq 0\}$ belongs to $\{2, 3, \infty\}$, classified as follows.*

(i) ***Class G** (Gaussian): $\Delta = 0$, $\alpha = 0$. $r_{\max} = 2$. Example: Heisenberg.*

(ii) ***Class L** (Lie): $\Delta = 0$, $\alpha \neq 0$. $r_{\max} = 3$. Example: affine Kac–Moody.*

(iii) ***Class M** (mixed): $\Delta \neq 0$. $r_{\max} = \infty$, and $S_r = \Delta \cdot R_r$ for universal rational functions $R_r \in \mathbb{Q}(\alpha, \kappa, S_4)$. Example: Virasoro, $\mathcal{W}_N$.*

*A fourth class, **class C** (contact, $r_{\max} = 4$), arises by stratum separation: the quartic contact invariant lives on a charged stratum that exits the single-line complex. Example: $\beta\gamma$.*

The shadow obstruction tower also carries a logarithmic connection.

Theorem 1.3 (Shadow connection; Theorem 8.1). *The shadow metric $Q(t)$ defines a logarithmic connection $\nabla^{\text{sh}} = d - \frac{Q'(t)}{2Q(t)}\,dt$ on $L \setminus \{Q = 0\}$, with regular singularities of residue $\frac{1}{2}$ at the zeros of Q. The monodromy around each branch point is -1. Under Koszul duality $A \mapsto A^!$, the connection transforms by the Verdier involution; for Virasoro:*

$$\Delta(\mathrm{Vir}_c) + \Delta(\mathrm{Vir}_{26-c}) \;=\; \frac{6960}{(5c+22)(152-5c)}.$$

Finally, the algebraicity theorem yields explicit genus-2 free energies by stable graph summation.

Theorem 1.4 (Genus-2 universality; Theorem 7.3). *For any chirally Koszul vertex algebra A with a single generator:*

$$F_2(A) \;=\; \kappa(A)\cdot\lambda_2^{\mathrm{FP}}, \qquad \lambda_2^{\mathrm{FP}} \;=\; \frac{7}{5760}.$$

For the $\mathcal{W}_3$ algebra with two generators T (weight 2) and W (weight 3), multi-generator universality at genus ≥ 2 is resolved negatively (Remark 7.1): the scalar formula $F_g = \kappa\cdot\lambda_g^{\mathrm{FP}}$ fails*, and the multi-channel stable graph sum over the seven genus-2 stable graphs yields a scalar part $F_2^{\mathrm{scal}} = (5c/6)\cdot 7/5760 = 7c/6912$ and a genuine cross-channel correction*

$$\delta F_2(\mathcal{W}_3) \;=\; \frac{c+204}{16c},$$

with per-graph contributions $3/c$ (banana), $9/(2c)$ (theta), $1/16$ (tadpole), and $21/(4c)$ (barbell). The tadpole contribution is independent of c.

1.4 Context

The algebraicity relation $H(t)^2 = t^4 Q(t)$ is structurally parallel to the loop equations of matrix models [1, 4] and to the Eynard–Orantin topological recursion [5], which compute higher-genus invariants from a spectral curve. In those frameworks the spectral curve is an input; here the spectral curve $\Sigma = \{H^2 = t^4 Q(t)\}$ is an *output* of the algebraicity theorem, determined by the vertex algebra A.

The shadow connection ∇^{sh} has Stokes data controlled by the critical discriminant: monodromy is trivial for classes G and L ($\Delta = 0$) and nontrivial for class M ($\Delta \neq 0$), in parallel with the wall-crossing formalism of Kontsevich–Soibelman [11]. The invariants live in the cyclic deformation complex of a vertex algebra, not in a motivic Hall algebra.

In classical Koszul duality, a Koszul algebra is determined by its quadratic dual $A^!$, and the bar resolution concentrates in degree 1. The chiral bar construction plays the same role, but the modular extension introduces genuinely new data: the shadow invariants S_r measure the complexity of higher-genus, higher-degree components of the bar differential, which have no classical analogue. The four-class partition (Theorem 1.2) classifies this complexity within the Koszul world; all four classes are Koszul.

1.5 Notation and conventions

Throughout, X denotes a smooth projective algebraic curve over a field k of characteristic zero. We work in the cohomological convention: all differentials have degree $+1$. The bar construction uses desuspension (s^{-1}, degree -1).

A *vertex algebra* is a vertex algebra in the sense of [7, 10], equipped with a conformal vector ω (the Virasoro element), a conformal grading $A = \bigoplus_{h\geq 0} A_h$ with $\dim A_h < \infty$, and the operator product expansion $a(z)b(w) \sim \sum_{n\geq 0}(a_{(n)}b)(w) \cdot (z-w)^{-n-1}$.

The *modular characteristic* $\kappa(A) \in \mathbb{Q}(c)$ is the genus-1 curvature scalar: the coefficient in $F_1(A) = \kappa(A)/24$. The *central charge* c is the standard Virasoro central charge. These are distinct invariants: $\kappa = k$ for the rank-1 Heisenberg at level k, $\kappa = c/2$ for Virasoro, and $\kappa = 5c/6$ for $\mathcal{W}_3$.

The Hodge class used in the scalar genus tower is the top Chern class

$$\lambda_g := c_g(\mathbb{E}_g) \in H^{2g}(\overline{\mathcal{M}}_g, \mathbb{Q}), \qquad \mathbb{E}_g = \pi_*\omega_\pi.$$

The first Chern class of the Hodge line is written $\lambda_1 = c_1(\det \mathbb{E}_g)$. The *$\hat{A}$-coefficients* λ_g^{FP} are the genus-g coefficients in the generating function $\sum_{g\geq 1} \lambda_g^{\mathrm{FP}} \hbar^{2g} = \frac{\hbar/2}{\sin(\hbar/2)} - 1$, so that $\lambda_1^{\mathrm{FP}} = 1/24$ and $\lambda_2^{\mathrm{FP}} = 7/5760$.

2 Chiral Koszul pairs and the bar construction

2.1 The chiral bar complex

Let A be a vertex algebra on a smooth projective curve X, equipped with a conformal vector and a positive-energy grading $A = \bigoplus_{h\geq 0} A_h$ with $\dim A_h < \infty$. The *Ran space* $\mathrm{Ran}(X) = \coprod_{n\geq 1} X^n/\Sigma_n$ parameterizes nonempty finite subsets of X. For each $n \geq 1$, the Fulton–MacPherson compactification $\mathrm{FM}_n(X)$ resolves the diagonals in X^n by a sequence of iterated real blowups along all partial diagonals, producing a smooth manifold-with-corners whose boundary strata are indexed by trees of collisions [8].

Definition 2.1 (Chiral bar complex)**.** The *chiral bar complex* $B(A)$ is the factorization coalgebra on $\mathrm{Ran}(X)$ with:

(i) *Underlying space*: the graded vector space $B(A)_n := (s^{-1}\overline{A})^{\otimes n}$, where $\overline{A} = A/k\mathbf{1}$ is the augmentation ideal and s^{-1} denotes desuspension (degree -1).

(ii) *Bar differential*: the coderivation $d_B \colon B(A)_n \to B(A)_n$ whose components are the collision residues

$$d_B|_{B(A)_n \to B(A)_{n-1}} = \sum_{i<j} \mathrm{Res}_{z_i \to z_j}\, a_i(z_i)\, a_j(z_j)\, d\log(z_i - z_j),$$

extracting the residue of the OPE along the logarithmic differential $d\log(z_i - z_j)$.

(iii) *Coproduct*: the cofactorization structure $\Delta \colon B(A)_n \to \bigoplus_{S\sqcup T=[n]} B(A)_{|S|} \otimes B(A)_{|T|}$ induced by the partition of points into clusters.

The identity $d_B^2 = 0$ encodes the associativity of the OPE.

Remark 2.2 (The $d\log$ measure). The bar differential extracts residues along $d\log(z-w)$, not along $dz/(z-w)$. The logarithmic differential absorbs one power of $(z-w)$: if the OPE has poles at $(z-w)^{-n-1}$, the collision residue has poles at $(z-w)^{-n}$. For Virasoro, the OPE has poles at orders 4, 2, 1; the bar r-matrix has poles at orders 3, 1 only.

Remark 2.3 (The ordered bar before averaging). The bar complex $B(A)$ of Definition 2.1 is the *symmetric* (factorization) bar $B^{\Sigma}(A)$, the Σ_n-coinvariant quotient of the ordered bar $B^{\mathrm{ord}}(A) = T^c(s^{-1}\overline{A})$. The ordered bar carries a deconcatenation coproduct (coassociative, not cocommutative) and supports the spectral r-matrix $r(z) \in \mathfrak{g}_A^{E_1}$; the shadow obstruction tower $\{S_r\}$ and the algebraicity relation $H(t)^2 = t^4 Q(t)$ are statements about the Σ_n-coinvariant image: the degree-2 scalar κ, given by $\mathrm{av}(r(z))$ in abelian and scalar families and by $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody, and its higher-degree extensions. The ordered bar is the level-2 chain object before averaging; the shadow tower is its averaged projection.

2.2 Chirally Koszul algebras

Definition 2.4 (Chirally Koszul)**.** A vertex algebra A is *chirally Koszul* if the bar cohomology $H^\bullet(B(A), d_B)$ is concentrated in bar degree 1: for every $n \geq 2$, the map $H^\bullet(B(A)_n) \to H^\bullet(B(A)_1)^{\otimes n}$ induced by the coproduct is a quasi-isomorphism.

Chirally Koszul algebras are the vertex-algebraic analogues of Koszul algebras in the sense of Priddy: the bar resolution has a minimal model. The standard families (Heisenberg, affine Kac–Moody at non-critical level, $\beta\gamma$, Virasoro, and $\mathcal{W}_N$) are all chirally Koszul (the universal affine vertex algebra $V_k(\mathfrak{g})$ is Koszul for all $k \neq -h^\vee$ by PBW concentration; the Virasoro and $\mathcal{W}_N$ algebras are Koszul by the same argument applied to their free-field realizations).

3 The modular convolution algebra and the Maurer–Cartan element

3.1 The modular cyclic deformation complex

Let A be a vertex algebra on a smooth projective curve X as in Section 2. The bar complex $B(A)$ carries, for each stable pair (g,n) with $2g-2+n>0$, a genus-g, n-marked component $B^{(g,n)}(A)$.

Definition 3.1 (Modular cyclic deformation complex)**.** The *modular cyclic deformation complex* of A is

$$\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A) \;:=\; \prod_{\substack{g,n \\ 2g-2+n>0}} \mathrm{CoDer}^{\mathrm{cyc}}\big(B^{(g,n)}(A)\big)[1],$$

where the product ranges over all stable pairs. The differential is $d = d_{\mathrm{int}} + d_{\mathrm{sew}}$:

(i) d_{int} is the internal cyclic coderivation differential on each component;

(ii) d_{sew} is the clutching operation induced by the boundary maps $\partial\colon \overline{\mathcal{M}}_{g,n} \supset \Delta_{g_1,S_1;\,g_2,S_2} \to \overline{\mathcal{M}}_{g_1,|S_1|+1} \times \overline{\mathcal{M}}_{g_2,|S_2|+1}$.

The Lie bracket is *graph composition*: for coderivations ξ and η at genera g_1, g_2 and degrees n_1, n_2,

$$[\xi,\eta] \;:=\; \sum_{\Gamma} \xi \circ_\Gamma \eta,$$

summed over stable graphs Γ connecting an output of ξ to an input of η with cross-polarization on internal edges.

The genus grading induces a complete descending filtration $F^\bullet \mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)$ with F^g consisting of elements supported at genus $\geq g$. At $g = 0$ the complex reduces to the ordinary cyclic deformation complex of A. The bracket respects the degree filtration: $[\xi, \eta]$ has degree $\leq n_1 + n_2 - 2$ (one input and one output are consumed by the internal edge).

3.2 The Maurer–Cartan element

The bar differential determines a canonical Maurer–Cartan element.

Proposition 3.2 (Bar-intrinsic MC element). *Let D_A denote the full bar differential on $B(A)$ (incorporating contributions from all genera and all boundary strata of $\overline{\mathcal{M}}_{g,n}$), and let d_0 denote its genus-0 linear part. Define*

$$\Theta_A \ := \ D_A - d_0 \ \in \ \mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A).$$

Then Θ_A satisfies the Maurer–Cartan equation

$$d_0 \Theta_A + \tfrac{1}{2}[\Theta_A, \Theta_A] = 0. \tag{1}$$

Proof. The identity $D_A^2 = 0$ is a formal consequence of $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$: the square of the bar differential decomposes into a sum over pairs of boundary strata whose contributions cancel by the codimension-2 face relation. Substituting $D_A = d_0 + \Theta_A$ into $D_A^2 = 0$ and expanding:

$$0 \ = \ d_0^2 + d_0 \Theta_A + \Theta_A \circ d_0 + \Theta_A \circ \Theta_A.$$

Since $d_0^2 = 0$ (the genus-0 bar differential squares to zero by OPE associativity) and $\Theta_A \circ d_0 + \Theta_A \circ \Theta_A = d_0 \Theta_A + \frac{1}{2}[\Theta_A, \Theta_A]$ in the cyclic coderivation Lie algebra, the result follows. □

3.3 The shadow obstruction tower

The MC element Θ_A encodes all genera and degrees. Its finite-order truncations form the shadow obstruction tower.

Definition 3.3 (Shadow obstruction tower). The *shadow obstruction tower* of A is the sequence of truncations $\Theta_A^{\leq r}$ $(r \geq 2)$ obtained by retaining only the components of degree at most r in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)$. The *degree-$r$ shadow* is the projection

$$\mathrm{Sh}_r(A) \ := \ \pi_r(\Theta_A) \ \in \ \mathrm{CoDer}^{\mathrm{cyc}}(B^{(\bullet, r)}(A))[1],$$

the component of Θ_A at degree exactly r.

The MC equation (1), projected onto the degree-r component, yields the *all-degree master equation*:

Proposition 3.4 (All-degree master equation). *For each $r \geq 2$, the degree-r projection of the MC equation gives*

$$d_0(\mathrm{Sh}_r) + \sum_{\substack{j+k=r+2 \\ j,k\geq 2}} [\mathrm{Sh}_j, \mathrm{Sh}_k] \ = \ 0.$$

In particular, Sh_r *is determined recursively by* $\mathrm{Sh}_2, \ldots, \mathrm{Sh}_{r-1}$ *together with the vanishing of a cohomological obstruction.*

Proof. The differential d_0 and the bracket $[-,-]$ on $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)$ both respect the degree filtration. Restricting the MC equation to degree r gives exactly the stated identity: the linear term is $d_0(\mathrm{Sh}_r)$, and the quadratic term $\frac{1}{2}[\Theta_A, \Theta_A]$ at degree r receives contributions from pairs (j,k) with $j+k=r+2$ (the bracket consumes two inputs via the internal edge, reducing total degree by 2). □

Definition 3.5 (Primary line and shadow data)**.** A *primary line* in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)$ is a one-dimensional subspace $L = k \cdot \eta$ spanned by a single primary field η of definite conformal weight. On L, the degree-r shadow restricts to a scalar: $\mathrm{Sh}_r|_L = S_r \cdot x^r$, where x is the coordinate on L and $S_r = S_r(A) \in \mathbb{Q}(c)$.

The first three scalars are:

- $S_2 =: \kappa$, the *modular characteristic*: $F_1(A) = \kappa/24$.
- $S_3 =: \alpha$, the *cubic shadow*: the three-point collision residue on $\mathrm{FM}_3(X)$.
- S_4, the *quartic shadow*: the four-point residue encoding the first nonlinear OPE invariant.

3.4 The shadow metric

Definition 3.6 (Shadow metric and critical discriminant)**.** Let L be a primary line with shadow data S_r, curvature $\kappa = S_2 \neq 0$, and cubic coefficient $\alpha = S_3$. Define the *weighted initial data* $a_0 := 2\kappa$, $a_1 := 3\alpha$, $a_2 := 4S_4$. The *shadow metric* on L is the quadratic polynomial

$$Q(t) \;:=\; a_0^2 + 2a_0a_1\,t + (a_1^2 + 2a_0a_2)\,t^2 \;=\; 4\kappa^2 + 12\kappa\alpha\,t + (9\alpha^2 + 16\kappa S_4)\,t^2.$$

The *critical discriminant* of L is

$$\Delta \;:=\; 8\kappa S_4 \;=\; a_0a_2.$$

Lemma 3.7 (Gaussian decomposition)**.**

$$Q(t) \;=\; (2\kappa + 3\alpha\,t)^2 \;+\; 2\Delta\,t^2.$$

The first term $(a_0 + a_1t)^2$ *is the shadow metric of the truncated tower with* $S_r = 0$ *for* $r \geq 4$*. The second term* $2\Delta\,t^2$ *is the interaction correction.*

Proof. $(2\kappa + 3\alpha t)^2 = 4\kappa^2 + 12\kappa\alpha\,t + 9\alpha^2\,t^2$. Adding $2\Delta\,t^2 = 16\kappa S_4\,t^2$ recovers $Q(t)$. □

4 The Riccati algebraicity theorem

4.1 Statement

Theorem 4.1 (Riccati algebraicity)**.** *Let A be a chirally Koszul vertex algebra, L a primary line in* $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A)$ *with shadow data S_r, curvature $\kappa = S_2 \neq 0$, and cubic coefficient $\alpha = S_3$. Define the shadow metric $Q(t)$ and the weighted shadow generating function* $H(t) := \sum_{r\geq 2} r\,S_r\,t^r$*. Then*

$$H(t)^2 \;=\; t^4\,Q(t). \tag{2}$$

Equivalently,

$$H(t) \;=\; t^2\sqrt{Q(t)}.$$

Every shadow coefficient S_r with $r \geq 5$ is determined by the three genus-0 invariants (κ, α, S_4) via the closed form $S_r = \frac{1}{r}\,[t^{r-2}]\sqrt{Q(t)}$.

4.2 The single-line recursion

The Maurer–Cartan equation on L reduces to a quadratic convolution identity.

Lemma 4.2 (MC recursion on a primary line)**.** *Let $P := 1/\kappa$. On the primary line L, the master equation (Proposition 3.4) reduces to the recursion*

$$S_r \;=\; -\frac{P}{2r} \sum_{\substack{j+k=r+2\\ 3\leq j\leq k}} c_{jk}\, jk\, S_j\, S_k, \qquad c_{jk} = \begin{cases} 1 & j<k,\\ 1/2 & j=k,\end{cases}$$

for all $r \geq 5$.

Proof. On the one-dimensional subspace L, the bracket $[\mathrm{Sh}_j, \mathrm{Sh}_k]$ restricts to a scalar pairing. The cyclic coderivation bracket on L evaluates as $[\mathrm{Sh}_j, \mathrm{Sh}_k]_L = c_{jk}\, jk\, P\, S_j S_k\, x^{j+k-2}$: the factor jk comes from the number of ways to select one input from each coderivation, the factor $P = 1/\kappa$ comes from the propagator on the internal edge (the inverse of the curvature pairing), and c_{jk} corrects for the symmetry when $j = k$. The linear term $d_0(\mathrm{Sh}_r)$ contributes $2r\,\kappa\, S_r$ (the eigenvalue of the curvature operator on the degree-r component). Dividing through by $2r\,\kappa$ gives the stated recursion.

Since $j \geq 3$ forces $k = r + 2 - j \leq r - 1$, the right-hand side involves only $S_3, \ldots, S_{r-1}$, and the recursion is well-defined. □

4.3 Proof of algebraicity

Proof of Theorem 4.1. Define $a_n := (n+2)\, S_{n+2}$ for $n \geq 0$, so that $a_0 = 2\kappa$, $a_1 = 3\alpha$, $a_2 = 4S_4$, and $H(t) = \sum_{n\geq 0} a_n\, t^{n+2}$. Set $F(t) := H(t)/t^2 = \sum_{n\geq 0} a_n\, t^n$. The claim (2) is equivalent to $F(t)^2 = Q(t)$.

Expand $F(t)^2 = \sum_{m\geq 0}\big(\sum_{i+j=m} a_i a_j\big)\, t^m$. The first three coefficients are:

$$\begin{aligned} m = 0:& \qquad a_0^2 = 4\kappa^2,\\ m = 1:& \qquad 2a_0 a_1 = 12\kappa\alpha,\\ m = 2:& \qquad a_1^2 + 2a_0 a_2 = 9\alpha^2 + 16\kappa S_4. \end{aligned}$$

These are exactly the coefficients of $Q(t)$ (Definition 3.6).

For $m \geq 3$, we must show that the convolution $\sum_{i+j=m} a_i a_j = 0$. Setting $r = m+2$, the convolution reads

$$\sum_{i+j=m} a_i a_j \;=\; \sum_{\substack{i+j=r-2\\ i,j\geq 0}} (i+2)(j+2)\, S_{i+2}\, S_{j+2} \;=\; \sum_{\substack{p+q=r+2\\ p,q\geq 2}} pq\, S_p\, S_q,$$

where $p = i + 2$, $q = j + 2$. Splitting off the terms involving $S_2 = \kappa$: the sum over $p + q = r + 2$ with $p, q \geq 2$ decomposes as the $p = 2$ term (contributing $2r\,\kappa\,S_r$), the $q = 2$ term (contributing $2r\,S_r\,\kappa$ again), and the sum with $p, q \geq 3$. So:

$$\sum_{i+j=m} a_i a_j \;=\; 4r\,\kappa\,S_r \;+\; \sum_{\substack{p+q=r+2\\ p,q\geq 3}} pq\,S_p\,S_q. \tag{3}$$

The MC recursion (Lemma 4.2) gives

$$S_r = -\frac{1}{2r\kappa} \sum_{\substack{p+q=r+2\\ 3\leq p\leq q}} c_{pq}\,pq\,S_p\,S_q.$$

Writing the symmetrized sum: $\sum_{\substack{p+q=r+2\\ p,q\geq 3}} pq\,S_p\,S_q \;=\; 2\sum_{\substack{p+q=r+2\\ 3\leq p<q}} pq\,S_pS_q \;+\; \sum_{\substack{p+q=r+2\\ p=q}} p^2\,S_p^2 \;=\; 2\sum_{\substack{p+q=r+2\\ 3\leq p\leq q}} c_{pq}\,pq\,S_pS_q = -4r\,\kappa\,S_r$. Substituting into (3):

$$\sum_{i+j=m} a_i a_j \;=\; 4r\,\kappa\,S_r + (-4r\,\kappa\,S_r) \;=\; 0.$$

Hence $[t^m](F^2 - Q) = 0$ for all $m \geq 3$. Combined with the verification at $m = 0, 1, 2$, this proves $F(t)^2 = Q(t)$, hence $H(t)^2 = t^4\,Q(t)$.

The closed form for S_r follows: $r\,S_r = [t^{r-2}]F(t) = [t^{r-2}]\sqrt{Q(t)}$. □

Remark 4.3 (Riccati ODE reformulation). Set $U(t) := \sum_{r\geq 3} S_r\,t^r$, the nonlinear part of the shadow generating function (removing Sh_2). The algebraic relation (2) is equivalent to the first-order ODE

$$2t\,U'(t) \;+\; \frac{1}{2\kappa}\big(U'(t)\big)^2 \;=\; 6\alpha\,t^3 + \frac{9\alpha^2 + 2\Delta}{2\kappa}\,t^4,$$

which is *quadratic in U'*. This is the precise sense in which the shadow obstruction tower is algebraic of degree 2: the generating function satisfies a Riccati-type equation, and all terms at t^r for $r \geq 5$ cancel identically. The ODE determines U uniquely from (κ, α, S_4) by power-series inversion.

4.4 The shadow connection

Definition 4.4 (Shadow connection)**.** The *shadow connection* on $L \setminus \{Q = 0\}$ is

$$\nabla^{\mathrm{sh}} \;:=\; d \;-\; \frac{Q'(t)}{2\,Q(t)}\,dt \;=\; d - d\Big(\log\sqrt{Q}\Big).$$

Proposition 4.5 (Properties of the shadow connection)**.**

(i) Flatness. $(\nabla^{\mathrm{sh}})^2 = 0$, *since $Q'/(2Q)\,dt$ is closed on a one-dimensional base.*

(ii) Regular singularities. *At each zero t_0 of Q, the connection form has a simple pole with residue $\frac{1}{2}$: near t_0, $Q(t) \approx Q'(t_0)(t - t_0)$, so $\frac{Q'}{2Q}\,dt \approx \frac{1}{2(t-t_0)}\,dt$.*

(iii) Koszul monodromy. *The monodromy around each branch point is* $\exp(2\pi i \cdot \frac{1}{2}) = -1$. *The monodromy representation factors through $\mathbb{Z}/2$; the nontrivial element acts by* -1.

(iv) Flat section. *The unique flat section with* $\Phi(0) = 1$ *is*

$$\Phi(t) \;=\; \sqrt{Q(t)/Q(0)} \;=\; \frac{\sqrt{Q(t)}}{2\kappa}.$$

The shadow generating function is the flat section weighted by the degree offset:

$$H(t) \;=\; 2\kappa\, t^2\, \Phi(t).$$

Proof. Parts (i)–(iii) are standard properties of the Gauss–Manin connection of the family $\{\sqrt{Q(t)}\}_{t\in L}$: the unique flat connection for which $\sqrt{Q}$ is a horizontal section. The residue computation at a simple zero of Q is immediate from the Laurent expansion. The monodromy is that of a square root around a simple branch point.

Part (iv): $\nabla^{\mathrm{sh}}\Phi = d\Phi - \frac{Q'}{2Q}\Phi\, dt = 0$ is verified directly, since $d\Phi = \frac{Q'}{2\sqrt{Q\cdot Q(0)}}\, dt = \frac{Q'}{2Q}\,\Phi\, dt$. The identity $H = 2\kappa\, t^2\, \Phi$ is Theorem 4.1: $H = t^2\sqrt{Q} = t^2 \cdot 2\kappa \cdot \Phi$. □

Remark 4.6 (Parallel transport of curvature). At $t = 0$, the shadow reduces to the curvature κ. The flat section $\Phi(t) = \sqrt{Q(t)}/(2\kappa)$ incorporates higher OPE data through ∇^{sh}. The generating function $H(t) = 2\kappa\, t^2\, \Phi(t)$ is *not* a flat section of ∇^{sh}: it has a double zero at $t = 0$ from the degree offset t^2.

Remark 4.7 (The Koszul monodromy). The shadow connection is the Gauss–Manin connection of a family of double covers: the spectral curve $\Sigma = \{H^2 = t^4 Q(t)\}$ is a double cover of the t-line, ramified at the zeros of Q. Analytic continuation of $\sqrt{Q}$ around a branch point returns $-\sqrt{Q}$; this is the standard $\mathbb{Z}/2$ monodromy of a square root.

Under Koszul duality $A \mapsto A^!$, the shadow data transforms: for the Virasoro family, $\kappa(c) \mapsto \kappa(26 - c)$, $\alpha(c) \mapsto \alpha(26 - c)$, $S_4(c) \mapsto S_4(26 - c)$. The shadow metric Q transforms by $c \mapsto 26 - c$, and the connection transforms by the Verdier involution. At $c = 13$ (the self-dual point): $Q = Q^!$ and the connection is self-dual.

5 The four-class classification

The algebraicity theorem (Theorem 4.1) implies that the shadow obstruction tower is determined by three invariants (κ, α, S_4), and that the generating function $H(t) = t^2\sqrt{Q(t)}$ is algebraic of degree 2 over the parameter line. The depth of the tower, defined as $r_{\max} := \sup\{r : S_r \neq 0\}$, is controlled entirely by the factorization of the shadow metric Q.

Theorem 5.1 (Single-line dichotomy). *On any primary line L with $\kappa \neq 0$ and critical discriminant $\Delta = 8\kappa S_4$, the shadow depth satisfies $r_{\max} \in \{2, 3, \infty\}$, classified as follows.*

(i) $\Delta = 0$, $\alpha = 0$*:* $Q = (2\kappa)^2$*;* $H(t) = 2\kappa\, t^2$*;* $r_{\max} = 2$ (class **G**, Gaussian).

(ii) $\Delta = 0$, $\alpha \neq 0$*:* $Q = (2\kappa + 3\alpha\, t)^2$*;* $H(t) = t^2(2\kappa + 3\alpha\, t)$*;* $r_{\max} = 3$ (class **L**, Lie).

(iii) $\Delta \neq 0$*: Q is irreducible over $\mathbb{Q}(\kappa, \alpha, S_4)$; $S_r \neq 0$ for all $r \geq 4$ generically;* $r_{\max} = \infty$ (class **M**, mixed).

For $r \geq 4$: $S_r = \Delta \cdot R_r$ with $R_r \in \mathbb{Q}(\alpha, \kappa, S_4)$ independent of the algebra family.

Proof. The Gaussian decomposition $Q(t) = (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2$ shows that Q is a perfect square if and only if $\Delta = 0$.

Case $\Delta = 0$. Then $Q = (2\kappa + 3\alpha t)^2$, so $H(t) = t^2(2\kappa + 3\alpha t)$. If also $\alpha = 0$, then $H(t) = 2\kappa t^2$, i.e., $S_r = 0$ for all $r \geq 3$. This is class **G**. If $\alpha \neq 0$, then H is a cubic polynomial in t, so $S_r = 0$ for $r \geq 4$ while $S_3 = \alpha \neq 0$. This is class **L**.

Case $\Delta \neq 0$. Then Q has two distinct zeros $t_\pm = \left(-6\kappa\alpha \pm \sqrt{-8\kappa^2\Delta}\right)/(9\alpha^2 + 2\Delta)$, so $\sqrt{Q(t)}$ is not polynomial. The Taylor coefficients $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q(t)}$ are nonzero for all $r \geq 4$ generically (the only way $[t^n]\sqrt{Q}$ can vanish identically is if $\sqrt{Q}$ is polynomial, which requires $\Delta = 0$). Hence $r_{\max} = \infty$: class **M**.

To see that S_r is divisible by Δ for $r \geq 4$, write $\sqrt{Q} = |2\kappa + 3\alpha t|\left(1 + 2\Delta t^2/(2\kappa + 3\alpha t)^2\right)^{1/2}$ and expand the binomial series. Every term beyond zeroth order contributes a factor of Δ, and all Taylor coefficients of H at order ≥ 4 come from these terms. The universal rational function R_r is the coefficient of Δ in this expansion and depends only on κ, α, and S_4. □

A fourth class escapes the single-line dichotomy.

Definition 5.2 (Class C: contact depth)**.** A chirally Koszul vertex algebra is of *class* **C** (contact, $r_{\max} = 4$) if

(a) on every one-dimensional primary line L, the shadow metric Q_L is a perfect square ($\Delta_L = 0$), so the single-line tower terminates at degree 3; but
(b) in the full cyclic deformation complex $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$, there exists a charged stratum supporting a nonzero quartic contact invariant $Q^{\mathrm{cont}} \neq 0$ at degree 4.

The quartic contact invariant lives on a two-dimensional weight/contact slice. Its self-bracket under the H-Poisson structure exits the single-line complex by *rank-one rigidity*: on a one-dimensional target, a quadratic self-bracket produces a scalar multiple of the generator, but the contact direction is orthogonal to the primary line, so the image lies outside the complex. The quintic obstruction o_5 therefore vanishes, and the tower terminates at degree 4.

Example 5.3 (Class C: the $\beta\gamma$ system)**.** The $\beta\gamma$ system of conformal weight λ has $\kappa = 6\lambda^2 - 6\lambda + 1$ (equal to $c/2$ where $c = 2(6\lambda^2 - 6\lambda + 1)$). On the weight-changing primary line, $\alpha = 0$ and $S_4 = 0$ (abelian OPE along this direction), so $\Delta = 0$ and the single-line tower terminates. But the full two-variable deformation complex contains a charged stratum with a nonzero quartic contact invariant. The quintic obstruction vanishes by rank-one rigidity. Hence $\beta\gamma$ is class **C** with $r_{\max} = 4$.

Remark 5.4 (The four classes partition the Koszul world). Every chirally Koszul vertex algebra with $\kappa \neq 0$ falls into exactly one of the four classes G, L, C, M. All four classes are Koszul: shadow depth classifies the complexity of the OPE within the Koszul world, not Koszulness itself. The four classes have the following structural characterization:

Class	α	Δ	$r_{\max}$	Examples
G	0	0	2	Heisenberg, lattice VOAs, free fermion
L	$\neq 0$	0	3	Affine Kac–Moody (all types)
C	0^*	0^*	4	$\beta\gamma$, bc ghosts
M	$\neq 0$	$\neq 0$	∞	Virasoro, $\mathcal{W}_N$ $(N \geq 3)$

The * for class **C** indicates that $\alpha = 0$ and $\Delta = 0$ hold on every one-dimensional primary line, while the quartic contact invariant lives on a higher-dimensional stratum.

5.1 Shadow depth and the depth decomposition

For algebras of class **M**, the shadow obstruction tower is infinite. The total depth admits a decomposition into arithmetic and algebraic components.

Definition 5.5 (Shadow depth decomposition). For a chirally Koszul vertex algebra A of class **M**, define the *total shadow depth*

$$d(A) \;=\; 1 + d_{\text{arith}} + d_{\text{alg}}$$

where:

- $d_{\text{alg}} \in \{0, 1, 2, \infty\}$ is the *algebraic depth*, measuring the complexity of the OPE-generated shadow obstruction tower on the primary line. By the algebraicity theorem, $d_{\text{alg}} = 0$ for all class **M** algebras: three invariants determine the full tower.
- $d_{\text{arith}} \geq 0$ is the *arithmetic depth*, counting contributions from cusp forms. Shadow coefficients of weight $\leq 2r$ at degree r lie in the subspace spanned by Eisenstein series; cusp-form contributions first appear at genus g where $S_{2g}(\mathrm{SL}_2(\mathbb{Z})) \neq 0$, i.e., at $g \geq 6$ (the Ramanujan Δ form at weight 12).

Remark 5.6. The algebraic depth is determined by genus-0 OPE data (Theorem 4.1). The arithmetic depth reflects automorphic phenomena in the genus expansion: cusp-form contributions are invisible at low genus but potentially enter at $g \geq 6$. For all families treated in Section 6, $d_{\text{arith}} = 0$ through the genera computed here.

5.2 Shadow radius and convergence

For class **M** algebras, the shadow obstruction tower is infinite. The algebraicity theorem determines the growth rate of the coefficients S_r exactly.

Definition 5.7 (Shadow growth rate). Let A be a chirally Koszul vertex algebra of class **M** with shadow data (κ, α, S_4) and $\Delta = 8\kappa S_4 \neq 0$. The *shadow growth rate* (or shadow radius) is

$$\rho(A) \;:=\; \frac{\sqrt{9\alpha^2 + 2\Delta}}{2|\kappa|}.$$

Proposition 5.8 (Shadow growth rate as reciprocal convergence radius). *The growth rate $\rho(A)$ equals the reciprocal of the radius of convergence of the weighted generating function $H(t) = t^2\sqrt{Q(t)}$:*

$$\rho(A) \;=\; \frac{1}{R}, \qquad R \;=\; \min_{Q(t_0)=0} |t_0|.$$

The branch points of H are the zeros of Q. When $\Delta > 0$ (the generic case for class **M***), these form a complex conjugate pair of equal modulus, so*

$$R \;=\; \frac{2|\kappa|}{\sqrt{9\alpha^2 + 2\Delta}}.$$

The shadow coefficients satisfy the asymptotic formula

$$S_r \;\sim\; C(A)\,\rho(A)^r\,r^{-5/2}\,\cos(r\theta + \phi), \qquad r \to \infty,$$

where θ and ϕ are determined by the argument of the nearest branch point, and $C(A)$ is a computable algebraic constant.

Proof. Since $H(t) = t^2\sqrt{Q(t)}$ and Q is quadratic, H is an algebraic function of degree 2 with branch points at the zeros of Q. The radius of convergence of the Taylor series of $\sqrt{Q}$ equals the distance from $t = 0$ to the nearest branch point. The zeros of Q are at $t_\pm = (-6\kappa\alpha \pm \sqrt{-8\kappa^2\Delta})/(9\alpha^2 + 2\Delta)$. For $\Delta > 0$, the discriminant $-8\kappa^2\Delta < 0$, so $t_\pm$ are complex conjugate with common modulus $|t_\pm| = 2|\kappa|/\sqrt{9\alpha^2 + 2\Delta}$. The transfer theorem of Flajolet–Sedgewick gives the stated asymptotic formula with exponent $-5/2$ (from the square-root singularity of an algebraic function of degree 2). □

Example 5.9 (Virasoro shadow growth rate). For Virasoro at central charge $c > 0$ with $\kappa = c/2$, $\alpha = 2$, $\Delta = 40/(5c + 22)$:

$$\rho(\mathrm{Vir}_c) \;=\; \frac{1}{c}\sqrt{\frac{180c + 872}{5c + 22}}.$$

The equation $\rho(\mathrm{Vir}_{c^*}) = 1$ is equivalent to the cubic

$$5c^3 + 22c^2 - 180c - 872 \;=\; 0,$$

whose unique positive real root is $c^* \approx 6.125$.

- For $c > c^*$: $\rho < 1$, the shadow obstruction tower converges (summable amplitudes). In particular, $\rho(13) \approx 0.467$, $\rho(26) \approx 0.232$.
- For $c < c^*$: $\rho > 1$, the shadow obstruction tower diverges (exponential growth). In particular, $\rho(1) \approx 6.242$, $\rho(1/2) \approx 12.532$.

All unitary minimal models ($c < 1$) have strongly divergent shadow obstruction towers. The string-theoretic value $c = 26$ has a convergent tower with $\rho \approx 0.232$.

6 Explicit shadow obstruction towers for the standard landscape

The shadow invariants $(\kappa, \alpha, S_4, \Delta, \rho)$ for all standard families are as follows.

6.1 Class G: Gaussian algebras

Proposition 6.1 (Heisenberg shadow obstruction tower). *For the Heisenberg vertex algebra $\mathcal{H}_k$ of rank 1 at level $k \neq 0$:*

$$\kappa \;=\; k, \qquad \alpha \;=\; 0, \qquad S_r \;=\; 0 \quad \textit{for all } r \geq 3.$$

The shadow obstruction tower terminates immediately: $\Theta^{\leq 2}_{\mathcal{H}_k} = k\,\eta^{\otimes 2}$, $Q(t) = 4k^2$ (constant), $H(t) = 2k\,t^2$. The genus-g formal scalar coefficient is $F_g(\mathcal{H}_k) = k \cdot \lambda_g^{\mathrm{FP}}$ for all $g \geq 1$.

Proof. The Heisenberg OPE $J(z)J(w) \sim k/(z-w)^2$ is purely quadratic: all higher collision residues on $\mathrm{FM}_n(X)$ for $n \geq 3$ vanish identically. Hence $S_r = 0$ for $r \geq 3$, $\alpha = 0$, $S_4 = 0$, $\Delta = 0$. The shadow metric is the constant $Q = 4k^2$, which is a perfect square. Class **G**. □

Proposition 6.2 (Lattice VOA shadow obstruction tower). *For the lattice vertex algebra V_Λ associated to an even positive-definite lattice Λ of rank r:*

$$\kappa \;=\; r, \qquad \alpha \;=\; 0, \qquad S_n \;=\; 0 \quad \textit{for all } n \geq 3.$$

Class **G**, *independent of the lattice cocycle.*

Proof. The primary line of a lattice VOA is spanned by the Heisenberg currents J^i, $i = 1, \dots, r$. The OPE $J^i(z)J^j(w) \sim \delta^{ij}/(z-w)^2$ is abelian (no cubic or higher terms on the primary line). The lattice vertex operators e^α have nontrivial OPEs but do not contribute to the primary-line shadow obstruction tower: they live on charged strata. The modular characteristic $\kappa = r$ equals the rank, independent of the specific lattice structure (even self-dual, Niemeier, Leech, etc.). □

Remark 6.3 (Free fermion). The free fermion ψ with $\psi(z)\psi(w) \sim 1/(z-w)$ has $c = 1/2$ and $\kappa = c/2 = 1/4$. On the single-generator primary line, the OPE is purely quadratic (Clifford), so $\alpha = 0$, $S_4 = 0$, $\Delta = 0$. Class **G**, $r_{\max} = 2$.

6.2 Class L: Lie algebras

Proposition 6.4 (Affine Kac–Moody shadow obstruction tower)**.** *For the universal affine vertex algebra $V_k(\mathfrak{g})$ at level $k \neq -h^\vee$ with $\mathfrak{g}$ simple of dimension $\dim(\mathfrak{g})$ and dual Coxeter number $h^\vee$:*

$$\kappa = \frac{\dim(\mathfrak{g})\,(k+h^\vee)}{2h^\vee}, \qquad \alpha \neq 0, \qquad S_4 = 0, \qquad \Delta = 0.$$

The cubic shadow α is nonzero (generated by the Lie bracket), and the quartic shadow S_4 vanishes on the primary line (killed by the Jacobi identity). The shadow obstruction tower terminates at degree 3: $r_{\max} = 3$, class **L***. The shadow metric is the perfect square $Q(t) = (2\kappa + 3\alpha\, t)^2$.*

Proof. The quartic collision residue on $\mathrm{FM}_4(X)$, restricted to the single-generator primary line, is proportional to $\sum_{\text{cyclic}} f^{abx}f^{xcd}$, which vanishes by the Jacobi identity $[[\cdot,\cdot],\cdot] + \text{cyclic} = 0$. Hence $S_4 = 0$, $\Delta = 8\kappa \cdot 0 = 0$, and Q is a perfect square. The cubic shadow α encodes the Lie bracket structure constants (the Killing 3-cocycle on $\mathfrak{g}$) and is nonzero on generic primary directions. □

The following table gives κ for the simple Lie algebras at level $k = 1$:

$\mathfrak{g}$	$\dim(\mathfrak{g})$	$h^\vee$	$\kappa(k)$	$\kappa(1)$	$c(1)$
$\mathfrak{sl}_2$	3	2	$\frac{3(k+2)}{4}$	$\frac{9}{4}$	1
$\mathfrak{sl}_3$	8	3	$\frac{4(k+3)}{3}$	$\frac{16}{3}$	2
$\mathfrak{sl}_N$	N^2-1	N	$\frac{(N^2-1)(k+N)}{2N}$	--	--
$\mathfrak{so}_5$	10	3	$\frac{10(k+3)}{6}$	$\frac{20}{3}$	$\frac{5}{2}$
G_2	14	4	$\frac{14(k+4)}{8}$	$\frac{35}{4}$	$\frac{14}{5}$
E_8	248	30	$\frac{248(k+30)}{60}$	$\frac{1922}{15}$	8

For all types and all non-critical levels $k \neq -h^\vee$: $\alpha \neq 0$, $S_4 = 0$, $\Delta = 0$, class **L**. The Koszul dual $V_k(\mathfrak{g})^! = \mathrm{CE}^{\mathrm{ch}}(\mathfrak{g}_{-k-2h^\vee})$ has modular characteristic $\kappa(V_k(\mathfrak{g})^!) = -\kappa(V_k(\mathfrak{g}))$, so $\kappa(k) + \kappa(k') = 0$ (exact anti-symmetry). Note: $V_k(\mathfrak{g})^!$ is the chiral CE algebra, not $V_{-k-2h^\vee}(\mathfrak{g})$ itself.

6.3 Class C: contact algebras

Proposition 6.5 ($\beta\gamma$ shadow obstruction tower). *For the $\beta\gamma$ system of conformal weight λ:*

$$\begin{aligned} \kappa &= 6\lambda^2 - 6\lambda + 1 = \tfrac{c}{2}, && c = 2(6\lambda^2 - 6\lambda + 1), \\ \alpha &= 0 && \text{(weight-changing line)}, \\ S_4 &= 0 && \text{(weight-changing line)}, \\ \Delta &= 0 && \text{(weight-changing line)}. \end{aligned}$$

On the weight-changing primary line, the tower terminates at degree 2 (class **G** *restricted to this line). The overall classification is class* **C***: a nonzero quartic contact invariant lives on the charged stratum of the full two-variable deformation complex, and the quintic vanishes by rank-one rigidity.*

Remark 6.6 (bc ghosts). The bc ghost system of spin j is the fermionic mirror of $\beta\gamma$, with $\kappa_{bc} = -(6j^2 - 6j + 1)$. At $j = 2$ (the reparametrization ghosts): $\kappa = -13$, $c = -26$. Same class **C**.

6.4 Class M: mixed algebras

Proposition 6.7 (Virasoro shadow obstruction tower). *For the Virasoro vertex algebra at central charge $c \neq 0$, $c \neq -22/5$:*

$$\kappa = \frac{c}{2}, \qquad \alpha = 2, \qquad S_4 = \frac{10}{c(5c+22)}, \qquad \Delta = \frac{40}{5c+22}.$$

The shadow metric is

$$Q(t) = (c+6t)^2 + \frac{80\,t^2}{5c+22},$$

the tower is infinite (*class* **M**)*, and the shadow growth rate is $\rho = \frac{1}{c}\sqrt{(180c+872)/(5c+22)}$.*

Theorem 6.8 (Virasoro shadow coefficients). *The Virasoro shadow obstruction tower coefficients through degree* 10 *are:*

$$\begin{aligned} S_2 &= \frac{c}{2}, \\ S_3 &= 2, \\ S_4 &= \frac{10}{c(5c+22)}, \\ S_5 &= \frac{-48}{c^2(5c+22)}, \\ S_6 &= \frac{80(45c+193)}{3\,c^3(5c+22)^2}, \\ S_7 &= \frac{-2880(15c+61)}{7\,c^4(5c+22)^2}, \\ S_8 &= \frac{80(2025c^2+16470c+33314)}{c^5(5c+22)^3}, \end{aligned}$$

$$S_9 = \frac{-1280(2025c^2 + 15570c + 29554)}{3\,c^6(5c+22)^3},$$
$$S_{10} = \frac{256(91125c^3 + 1050975c^2 + 3989790c + 4969967)}{c^7(5c+22)^4}.$$

All coefficients belong to $\mathbb{Q}(c)$ with poles only at $c = 0$ and $c = -22/5$. The signs alternate starting from S_5: $S_5 < 0$, $S_6 > 0$, $S_7 < 0, \ldots$ for $c > 0$. The c-independence of $S_3 = 2$ is now proved algebraically (not merely computed): it follows from a direct residue extraction; see [14].

Proof. The computation proceeds inductively using the master equation. The Hessian propagator on the primary line is $P = 2/c$ (inverse of $\kappa = c/2$). The degree-r obstruction is

$$o_r = \sum_{\substack{j+k=r+2\\ j,k\geq 2}} \epsilon_{jk}\,\{S_j\,x^j,\ S_k\,x^k\}_H, \qquad \{f,g\}_H = f'(x)\,P\,g'(x),$$

where $\epsilon_{jk} = 1$ for $j \neq k$ and $\epsilon_{jk} = 1/2$ for $j = k$. The shadow coefficient is $S_r = -o_r/(2r)$, where we extract the coefficient of x^r.

Each coefficient agrees with the closed-form extraction $S_r = \frac{1}{r}[t^{r-2}]\sqrt{Q_{\mathrm{Vir}}(t)}$ from Theorem 4.1. $\square$

Evaluations at three central charges:

r	$S_r(c{=}1)$	$S_r(c{=}13)$	$S_r(c{=}26)$
2	$1/2$	$13/2$	13
3	2	2	2
4	$10/27$	$10/1131$	$5/1976$
5	$-16/9$	$-16/4901$	$-3/6422$
6	$19040/2187$	$62240/49887279$	$6815/76139232$
7	$-24320/567$	$-81920/168138607$	$-20295/1154778352$
8	$4144720/19683$	$47171920/244497554379$	$4576085/1303909727744$

At $c = 26$ the coefficients decay rapidly ($\rho \approx 0.232$); at $c = 13$ moderately ($\rho \approx 0.467$); at $c = 1$ they grow exponentially ($\rho \approx 6.242$).

Proposition 6.9 ($\mathcal{W}_N$ shadow data). *For the $\mathcal{W}_N$ algebra ($N \geq 3$) at central charge c, the modular characteristic on the T-line (Virasoro direction) is $\kappa_T = c/2$ with shadow data identical to Virasoro. The total modular characteristic is*

$$\kappa(\mathcal{W}_N) = \rho_N \cdot c, \qquad \rho_N := H_N - 1 = \sum_{j=2}^{N} \frac{1}{j},$$

where $H_N = \sum_{j=1}^{N} 1/j$ is the N-th harmonic number.

The anomaly ratios ρ_N for small N are:

N	$\mathcal{W}_N$	$\rho_N = H_N - 1$	$\kappa(\mathcal{W}_N)$
2	Virasoro	$1/2$	$c/2$
3	$\mathcal{W}_3$	$5/6$	$5c/6$
4	$\mathcal{W}_4$	$13/12$	$13c/12$
5	$\mathcal{W}_5$	$77/60$	$77c/60$
6	$\mathcal{W}_6$	$29/20$	$29c/20$
7	$\mathcal{W}_7$	$223/140$	$223c/140$

All $\mathcal{W}_N$ for $N \geq 3$ are class **M**: the T-line shadow data coincides with Virasoro ($\alpha_T = 2, \Delta_T = 40/(5c + 22)$), and the W-line carries independent class **M** data with even-degree shadow obstruction tower.

6.5 Master table

The following table collects the shadow invariants for the entire standard landscape.

Family	κ	α	S_4	Δ	Class	$r_{\max}$	ρ
$\mathcal{H}_k$ (Heisenberg)	k	0	0	0	**G**	2	0
V_Λ (lattice)	$\mathrm{rk}(\Lambda)$	0	0	0	**G**	2	0
Free fermion	$1/4$	0	0	0	**G**	2	0
$V_k(\mathfrak{sl}_2)$	$\frac{3(k+2)}{4}$	$\neq 0$	0	0	**L**	3	0
$V_k(\mathfrak{sl}_N)$	$\frac{(N^2-1)(k+N)}{2N}$	$\neq 0$	0	0	**L**	3	0
$V_k(\mathfrak{g})$ (general)	$\frac{\dim\mathfrak{g}\,(k+h^\vee)}{2h^\vee}$	$\neq 0$	0	0	**L**	3	0
$\beta\gamma_\lambda$	$6\lambda^2-6\lambda+1$	0^*	0^*	0^*	**C**	4	--
bc_j	$-(6j^2-6j+1)$	0^*	0^*	0^*	**C**	4	--
Vir_c	$c/2$	2	$\frac{10}{c(5c+22)}$	$\frac{40}{5c+22}$	**M**	∞	$\frac{1}{c}\sqrt{\frac{180c+872}{5c+22}}$
$\mathcal{W}_3$	$5c/6$	$\neq 0$	$\neq 0$	$\neq 0$	**M**	∞	--
$\mathcal{W}_N$ $(N\geq 3)$	$(H_N-1)\,c$	$\neq 0$	$\neq 0$	$\neq 0$	**M**	∞	--

The * entries for classes C indicate values on the weight-changing primary line; the quartic contact invariant lives on a higher-dimensional stratum. For class **M**, the dashes (--) indicate that the shadow radius depends on the choice of primary line and has not been computed in closed form for the full multi-channel problem.

6.6 Complementarity of discriminants

Koszul duality $A \mapsto A^!$ transforms the shadow invariants in a controlled way. For Virasoro, the chiral Koszul dual is $\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$ (the duality $c \mapsto 26 - c$).

Proposition 6.10 (Discriminant complementarity). *The critical discriminants satisfy*

$$\Delta(\mathrm{Vir}_c) + \Delta(\mathrm{Vir}_{26-c}) = \frac{6960}{(5c+22)(152-5c)}.$$

The numerator $6960 = 40 \times 174$ *is independent of* c*. At the self-dual point* $c = 13$*:* $\Delta(13) = 40/87$*, and the sum is* $80/87$*. The product* $(5c+22)(152-5c)$ *has maximum* 7569 *at* $c = 13$*.*

Proof. Direct computation:

$$\begin{aligned}\Delta(c) + \Delta(26-c) &= \frac{40}{5c+22} + \frac{40}{5(26-c)+22} \\ &= \frac{40}{5c+22} + \frac{40}{152-5c} \\ &= 40 \cdot \frac{(152-5c)+(5c+22)}{(5c+22)(152-5c)} = \frac{40 \times 174}{(5c+22)(152-5c)}.\end{aligned}$$

□

Proposition 6.11 (Shadow radius at special points). *The Virasoro shadow growth rate at selected central charges:*

c	$\rho(c)$	$\rho(26{-}c)$	*convergent?*	*interpretation*
$1/2$	12.53	0.224	*no/yes*	*Ising/*$\mathrm{Vir}_{51/2}$
1	6.242	0.242	*no/yes*	*free boson/*Vir_{25}
c^*	1	0.430	*marginal/yes*	*critical*
13	0.467	0.467	*yes/yes*	*self-dual*
25	0.242	6.242	*yes/no*	*dual of* $c{=}1$
26	0.232	∞	*yes/–-*	*string/*$\kappa^!{=}0$

At $c = 13$*, the shadow growth rates are equal (*$\rho = \rho^!$*), reflecting the self-duality of* Vir_{13}*. At* $c = 26$*, the Koszul dual* Vir_0 *has* $\kappa^! = 0$ *(the bar complex is uncurved), and* $\rho^!$ *is undefined.*

Remark 6.12 (Duality and convergence). The Koszul duality $c \leftrightarrow 26 - c$ exchanges convergent and divergent towers: if $\rho(c) < 1$ then $\rho(26-c) > 1$ generically, and vice versa. The self-dual point $c = 13$ is the unique central charge where $\rho(A) = \rho(A^!)$. This reflects the complementarity principle: the sum $\Delta(c) + \Delta(26-c) = 6960/[(5c+22)(152-5c)]$ is symmetric in $c \leftrightarrow 26 - c$, with maximum at $c = 13$.

7 The $\mathcal{W}_3$ cross-channel computation

The genus-2 free energy of a single-generator chirally Koszul algebra is determined by the modular characteristic alone: $F_2(A) = \kappa(A) \cdot \lambda_2^{\mathrm{FP}}$, where $\lambda_2^{\mathrm{FP}} = 7/5760$. For a multi-generator algebra, the graph sum acquires *cross-channel* contributions from edges carrying different generators. The $\mathcal{W}_3$ algebra is the simplest instance.

The cross-channel correction, computed by stable graph summation, is

$$\delta F_2(\mathcal{W}_3) = \frac{c+204}{16c}, \tag{4}$$

the first explicit multi-channel genus-2 amplitude for any vertex algebra with generators of distinct conformal weight.

Remark 7.1 (Status). The cross-channel correction (4) uses per-channel genus-1 vertex factors $\kappa_i/24$ without R-matrix corrections. The open problem op:multi-generator-universality is resolved negatively: the R-matrix dressing does *not* cancel the cross-channel correction at $g \geq 2$. The per-channel scalar part $F_2^{\mathrm{scal}} = (5c/6) \cdot 7/5760 = 7c/6912$ is unconditional; the cross-channel correction $\delta F_2 = (c+204)/(16c)$ is a genuine additional term.

7.1 The $\mathcal{W}_3$ Frobenius algebra

The $\mathcal{W}_3$ algebra has two strong generators: T (conformal weight 2, the Virasoro stress tensor) and W (conformal weight 3, the spin-3 current). The bar-complex CohFT has a two-dimensional Frobenius algebra with the following data.

Lemma 7.2 ($\mathcal{W}_3$ Frobenius data)**.**

(i) Curvature metric. $\eta_{TT} = \kappa_T = c/2$, $\eta_{WW} = \kappa_W = c/3$, $\eta_{TW} = 0$ (*diagonal: T and W are orthogonal channels*). *The total modular characteristic is* $\kappa(\mathcal{W}_3) = \kappa_T + \kappa_W = 5c/6$.

(ii) Propagators. $\eta^{TT} = 2/c$, $\eta^{WW} = 3/c$. *Both channels use the weight-*1 *bar propagator* $d\log E(z, w)$, *regardless of the conformal weight of the generator.*

(iii) Three-point structure constants. *The $\mathbb{Z}_2$ symmetry $W \mapsto -W$ forces all structure constants with an odd number of W indices to vanish. The nonzero constants are*

$$C_{TTT} = C_{TWW} = C_{WWT} = c.$$

In particular, $C_{TTW} = C_{WWW} = 0$.

(iv) Four-point vertex. *For the genus-*0 *quartic vertex in the s-channel factorization:*

$$V_{0,4}(i, i \mid j, j) \;=\; \sum_{m \in \{T,W\}} \eta^{mm}\, C_{iim}\, C_{jjm} \;=\; 2c$$

for all channel pairs (i, j). The universality follows because only the T-channel contributes: $\eta^{TT} C_{iiT} C_{jjT} = (2/c) \cdot c \cdot c = 2c$.

Proof. Part (i): the curvature κ_i is the leading OPE pole coefficient. From $T_{(3)}T = c/2$ and $W_{(5)}W = c/3$. Part (iii): from the bar-complex r-matrix (OPE poles shifted by -1 via $d\log$): $T_{(1)}T = 2T$ gives $C_{TT}^{T} = 2$, so $C_{TTT} = \eta_{TT} \cdot 2 = c$; $T_{(1)}W = 3W$ gives $C_{TW}^{W} = 3$, so $C_{TWW} = \eta_{WW} \cdot 3 = c$; $W_{(3)}W = 2T$ gives $C_{WW}^{T} = 2$, so $C_{WWT} = \eta_{TT} \cdot 2 = c$. Part (iv): direct computation. □

7.2 Stable graphs at genus 2

There are seven stable graphs of genus 2 with no marked points:

Graph	Vertices	Edges	$\lvert\mathrm{Aut}\rvert$	h^1
Γ_0: smooth	$1 \times (g{=}2, \mathrm{val} = 0)$	0	1	0
Γ_1: figure-eight	$1 \times (g{=}1, \mathrm{val} = 2)$	1 self-loop	2	1
Γ_2: banana	$1 \times (g{=}0, \mathrm{val} = 4)$	2 self-loops	8	2
Γ_3: dumbbell	$(g{=}1, 1) + (g{=}1, 1)$	1 bridge	2	0
Γ_4: theta	$(g{=}0, 3) + (g{=}0, 3)$	3 bridges	12	2
Γ_5: tadpole	$(g{=}0, 3) + (g{=}1, 1)$	1 self-loop + 1 bridge	2	1
Γ_6: barbell	$(g{=}0, 3) + (g{=}0, 3)$	2 self-loops + 1 bridge	8	2

Each graph satisfies the genus formula $g = h^1 + \sum_v g_v = 2$ and the stability condition $2g_v + \mathrm{val}_v \geq 3$ at every vertex.

Each edge carries a channel label $\sigma(e) \in \{T, W\}$. The amplitude of a graph Γ with channel assignment $\sigma\colon E(\Gamma) \to \{T, W\}$ is

$$A(\Gamma, \sigma) = \frac{1}{|\mathrm{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} \eta^{\sigma(e),\sigma(e)} \prod_{v \in V(\Gamma)} V_v(g_v, \text{ channels}_v),$$

where V_v is the vertex factor: C_{ijk} for genus-0 trivalent vertices, $V_{0,4}$ for genus-0 four-valent vertices, and $\kappa_i/24$ for genus-1 vertices with a single half-edge of channel i.

An assignment σ is *mixed* if not all edges carry the same channel. The cross-channel correction is the sum of mixed-assignment amplitudes over all graphs.

7.3 Graph-by-graph computation

We compute the mixed-channel contribution from each of the six boundary graphs (Γ_1–Γ_6; the smooth graph Γ_0 carries no edges).

Γ_1: figure-eight

One edge, one self-loop at the genus-1 vertex. The single edge carries either T or W; there is no mixed assignment. Cross-channel contribution: 0.

Γ_2: banana

Two self-loops at a genus-0 four-valent vertex, with $|\mathrm{Aut}| = 8$. The four channel assignments $(\sigma_1, \sigma_2) \in \{T, W\}^2$ and the quartic vertex universality $V_{0,4}(i, i \mid j, j) = 2c$ give:

$$\begin{aligned}
(T, T): &\quad \eta^{TT}\,\eta^{TT} \cdot 2c = (2/c)^2 \cdot 2c = 8/c, \\
(W, W): &\quad \eta^{WW}\,\eta^{WW} \cdot 2c = (3/c)^2 \cdot 2c = 18/c, \\
(T, W) \text{ or } (W, T): &\quad \eta^{TT}\,\eta^{WW} \cdot 2c = (2/c)(3/c) \cdot 2c = 12/c, \qquad \text{two assignments.}
\end{aligned}$$

The mixed raw amplitude is $2 \cdot 12/c = 24/c$. Dividing by $|\mathrm{Aut}| = 8$:

$$\delta_{\text{banana}} = \frac{24}{8c} = \frac{3}{c}. \tag{5}$$

Γ_3: dumbbell

One bridge connecting two genus-1 vertices. The single edge carries either T or W; the off-diagonal metric $\eta^{TW} = 0$ kills any notion of mixing. Cross-channel contribution: 0.

Γ_4: theta

Three bridges connecting two genus-0 trivalent vertices, with $|\mathrm{Aut}| = 12$. There are $2^3 = 8$ channel assignments.

The $\mathbb{Z}_2$ parity $C_{TTW} = C_{WWW} = 0$ kills every assignment in which either vertex receives an odd number of W half-edges. Since each vertex sees exactly one half-edge from each bridge, the constraint is that the number of W-labeled edges must be even.

- (T, T, T): all-T, diagonal. Amplitude: $(2/c)^3 \cdot c^2 = 8/c$. With 1/12: $2/(3c)$.
- (T, T, W) and permutations (3 assignments): each vertex gets (T, T, W). $C_{TTW} = 0$, so all three vanish.
- (T, W, W) and permutations (3 assignments): each vertex gets (T, W, W). $C_{TWW} = c$ at both vertices. Propagators: $\eta^{TT}\,(\eta^{WW})^2 = (2/c)(3/c)^2 = 18/c^3$. Raw: $18c^2/c^3 = 18/c$. Three such assignments; total mixed raw: $54/c$. With 1/12: $54/(12c) = 9/(2c)$.
- (W, W, W): $C_{WWW} = 0$, vanishes.

The mixed contribution from the theta graph is

$$\delta_{\text{theta}} = \frac{54}{12c} = \frac{9}{2c}.$$

Γ_5: tadpole

One self-loop at the genus-0 trivalent vertex v_0, one bridge from v_0 to the genus-1 vertex v_1. $|\mathrm{Aut}| = 2$. Write (σ_1, σ_2) for the channels of the self-loop and the bridge.

- (T, W): self-loop T, bridge W. v_0 gets half-edges (T, T, W): $C_{TTW} = 0$. Vanishes.
- (W, T): self-loop W, bridge T. v_0 gets half-edges (W, W, T): $C_{WWT} = c$. v_1 gets half-edge T: $V_{1,1}(T) = \kappa_T/24 = c/48$. Propagators: $\eta^{WW}\,\eta^{TT} = (3/c)(2/c) = 6/c^2$. Raw: $(6/c^2)\cdot c \cdot (c/48) = 1/8$. With $1/|\mathrm{Aut}| = 1/2$: 1/16.

The tadpole mixed amplitude is

$$\delta_{\text{tadpole}} = \frac{1}{16}.$$

This is *independent of the central charge c*: $(3/c)(2/c)\cdot c\cdot(c/48) = 1/8$, and 1/8 divided by $|\mathrm{Aut}| = 2$ is 1/16.

Γ_6: barbell

Two genus-0 trivalent vertices v_1, v_2, each carrying a self-loop, connected by a single bridge. $|\mathrm{Aut}| = 8$ (swapping $v_1 \leftrightarrow v_2$ and independently swapping the two half-edges of each self-loop).

Write $(\sigma_1, \sigma_2, \sigma_3)$ for the channels of the self-loop at v_1, the self-loop at v_2, and the bridge. Vertex v_i receives half-edges $(\sigma_i, \sigma_i, \sigma_3)$ and contributes the three-point function $C_{\sigma_i\sigma_i\sigma_3}$.

- (T, W, T): v_1 gets (T, T, T), $C_{TTT} = c$; v_2 gets (W, W, T), $C_{WWT} = c$. Propagators: $\eta^{TT}\,\eta^{WW}\,\eta^{TT} = (2/c)(3/c)(2/c) = 12/c^3$. Raw: $12c^2/c^3 = 12/c$.

- (W, T, T): by the $v_1 \leftrightarrow v_2$ symmetry, same raw amplitude $12/c$.
- (W, W, T): v_1 gets (W, W, T), $C_{WWT} = c$; v_2 gets (W, W, T), $C_{WWT} = c$. Propagators: $(\eta^{WW})^2\, \eta^{TT} = (3/c)^2(2/c) = 18/c^3$. Raw: $18c^2/c^3 = 18/c$.
- All other mixed assignments vanish: any assignment with bridge channel W requires $C_{\sigma_i\sigma_i W}$; $C_{TTW} = 0$ and $C_{WWW} = 0$.

Total mixed raw amplitude: $12/c + 12/c + 18/c = 42/c$. Dividing by $|\,\mathrm{Aut}\,| = 8$:

$$\delta_{\text{barbell}} = \frac{42}{8c} = \frac{21}{4c}. \tag{6}$$

7.4 The cross-channel correction

Theorem 7.3 ($\mathcal{W}_3$ cross-channel correction)**.** *The cross-channel correction to the genus-2 free energy of $\mathcal{W}_3$ is*

$$\delta F_2(\mathcal{W}_3) = \underbrace{\frac{3}{c}}_{banana} + \underbrace{\frac{9}{2c}}_{theta} + \underbrace{\frac{1}{16}}_{tadpole} + \underbrace{\frac{21}{4c}}_{barbell} = \frac{c+204}{16c}.$$

The open problem op:multi-generator-universality *is resolved negatively. The full genus-2 free energy decomposes as*

$$F_2(\mathcal{W}_3) = \underbrace{\frac{5c}{6} \cdot \frac{7}{5760}}_{F_2^{\text{scal}} = 7c/6912} + \underbrace{\frac{c+204}{16c}}_{\delta F_2^{\text{cross}}}.$$

Proof. The cross-channel contributions from all seven graphs are computed in (5)–(6). Four graphs contribute: banana ($3/c$), theta ($9/(2c)$), tadpole ($1/16$), barbell ($21/(4c)$). The remaining three (smooth, figure-eight, dumbbell) have zero mixed amplitude: the smooth graph has no edges, while the figure-eight and dumbbell each have a single edge, so every assignment is diagonal.

For the sum:

$$\frac{3}{c} + \frac{9}{2c} + \frac{1}{16} + \frac{21}{4c} = \frac{48}{16c} + \frac{72}{16c} + \frac{c}{16c} + \frac{84}{16c} = \frac{c+204}{16c}. \qquad \square$$

Remark 7.4 (Properties of the cross-channel correction).

(i) *The c-independent piece.* The tadpole contribution $1/16$ is the only c-independent term: $\eta^{WW}\, \eta^{TT}\, C_{WWT}\, (\kappa_T/24) = (3/c)(2/c) \cdot c \cdot (c/48) = 1/8$, and every factor of c cancels. As $c \to \infty$, $\delta F_2 \to 1/16$.

(ii) *The rational piece.* The c-dependent terms total $51/(4c)$: banana $3/c$, theta $9/(2c)$, and barbell $21/(4c)$. All three arise from genus-0 vertices, mediated by the $\mathbb{Z}_2$-allowed mixed assignments. They dominate at small c.

(iii) *Vanishing at $c = -204$.* The cross-channel correction vanishes at $c = -204$.

(iv) *Relative magnitude.* The ratio $\delta F_2/F_2^{\text{scal}} = 432(c+204)/(7c^2)$ tends to zero as $c \to \infty$: the cross-channel correction becomes subleading at large central charge. The crossover $\delta F_2 = F_2^{\text{scal}}$ occurs at $c \approx 147$. At moderate c the correction dominates: at $c = 50$, $\delta F_2/F_2^{\text{scal}} \approx 6.3$. At large c the correction is suppressed: at $c = 500$, $\delta F_2/F_2^{\text{scal}} \approx 0.17$.

Remark 7.5 (The $\mathbb{Z}_2$ selection rule). The $\mathbb{Z}_2$ symmetry $W \mapsto -W$ of the $\mathcal{W}_3$ algebra forces $C_{TTW} = C_{WWW} = 0$, eliminating every mixed assignment in which a vertex receives an odd number of W-legs. For the theta graph, this kills the three (T, T, W)-type assignments and preserves the three (T, W, W)-type assignments. For $\mathcal{W}_N$ with $N \geq 4$, the additional generators (spins $4, \ldots, N$) break this symmetry.

8 The shadow connection and Koszul duality

Theorem 8.1 (Shadow connection). *The shadow metric $Q(t)$ defines a logarithmic connection*

$$\nabla^{\mathrm{sh}} = d - \frac{Q'(t)}{2\,Q(t)}\,dt$$

on $L \setminus \{Q = 0\}$.

(i) The connection has regular singularities of residue $\frac{1}{2}$ at each zero of Q.

(ii) The monodromy around each branch point is -1.

(iii) The flat section with $\Phi(0) = 1$ is $\Phi(t) = \sqrt{Q(t)/Q(0)}$, and $H(t) = 2\kappa\, t^2\, \Phi(t)$.

(iv) Under $A \mapsto A^!$ (Koszul duality), for Virasoro:

$$\Delta(\mathrm{Vir}_c) + \Delta(\mathrm{Vir}_{26-c}) = \frac{6960}{(5c+22)(152-5c)}.$$

Proof. Parts (i)–(iii) are proved in Proposition 4.5. Part (iv) is Proposition 6.10. □

9 Outlook

Three directions extend the algebraicity theorem.

Shadow obstruction towers for non-standard families

The standard landscape (Heisenberg, affine Kac–Moody, $\beta\gamma$, Virasoro, $\mathcal{W}_N$) exhausts the four shadow depth classes G, L, C, M. Non-standard families (admissible-level quotients, $N = 2$ superconformal algebras, non-simply-laced affine vertex algebras at fractional level) pose new questions. For admissible-level quotients $L_k(\mathfrak{g})$, null vectors modify the bar complex and may shift the shadow depth class. The Bershadsky–Polyakov algebra $\mathcal{W}_3^{(2)}$ at admissible levels and the triplet $\mathcal{W}(p)$ algebras are concrete test cases. Whether the four-class partition exhausts all chirally Koszul algebras, or whether exotic depth classes (e.g., $r_{\max} = 5$) arise outside the standard landscape, is open.

The Pixton connection

For class **M** algebras (Virasoro, $\mathcal{W}_N$), the infinite shadow obstruction tower $\{S_r\}_{r\geq 2}$ produces an infinite sequence of tautological classes on $\overline{\mathcal{M}}_{g,n}$ via the Maurer–Cartan equation. At genus 2–3, computations in the modular convolution algebra yield tautological relations that belong to the Pixton ideal $\mathcal{I}_{\mathrm{Pix}} \subset R^*(\overline{\mathcal{M}}_{g,n})$, the ideal of tautological relations constructed by Pixton and proved to hold by Janda–Pandharipande–Pixton–Zvonkine [9] (completeness is conjectural in general). The MC equation $D^2 = 0$ forces algebraic relations among stable-graph amplitudes at each genus, and these project to tautological identities on $\overline{\mathcal{M}}_g$.

The conjecture is that the Virasoro MC tower generates the entire Pixton ideal, via the shadow-to-strata map sending each MC obstruction o_r to a codimension-$(r-2)$ tautological class. This is verified at genus 2 (the planted-forest formula $\delta^{(2,0)}_{\mathrm{pf}} = S_3(10S_3 - \kappa)/48$ lies in the strata algebra) and at genus 3 (the 11-term polynomial in κ, S_3, S_4, S_5 matches known relations). The main obstruction is the flat unit question for the shadow CohFT, needed to apply Givental–Teleman reconstruction.

Analytic completion and shadow convergence

The shadow growth rate $\rho(A)$ (Definition 5.7) divides the Virasoro family into a convergent regime ($c > c^*$, $\rho < 1$) and a divergent regime ($c < c^*$, $\rho > 1$). The convergent shadow obstruction tower defines a shadow partition function $Z^{\mathrm{sh}}(q,t)$ with absolute convergence in both the genus variable q and the degree variable t. The string partition function $Z^{\mathrm{str}}(g_s) = \sum_g F_g\, g_s^{2g-2}$ diverges factorially as $(2g)!$ by Shenker's argument.

The shadow obstruction tower converges because its coefficients decay as $S_r \sim \rho^r\, r^{-5/2}$ (algebraic, from a square-root singularity), while string amplitudes grow as $(2g)!$ (factorial, from the volume of $\overline{\mathcal{M}}_g$). The shadow partition function is therefore a resummation of the factorially divergent string expansion, converging on a domain determined by the shadow radius.

The passage from algebraic shadow obstruction tower to analytic partition function requires the sewing envelope construction: a Hausdorff completion of the vertex algebra in the all-amplitude seminorm topology. The Hilbert–Schmidt sewing condition (that the sewing operator has finite Hilbert–Schmidt norm) is satisfied for the entire standard landscape and implies trace-class closed amplitudes at all genera. The resulting analytic bar coalgebra lives in the coderived category at genus $g \geq 1$, where curvature obstructs the ordinary derived category.

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